

Assignment 02

Parametric Pavillion

Course: Arch 565 - Advanced Computer Applications II

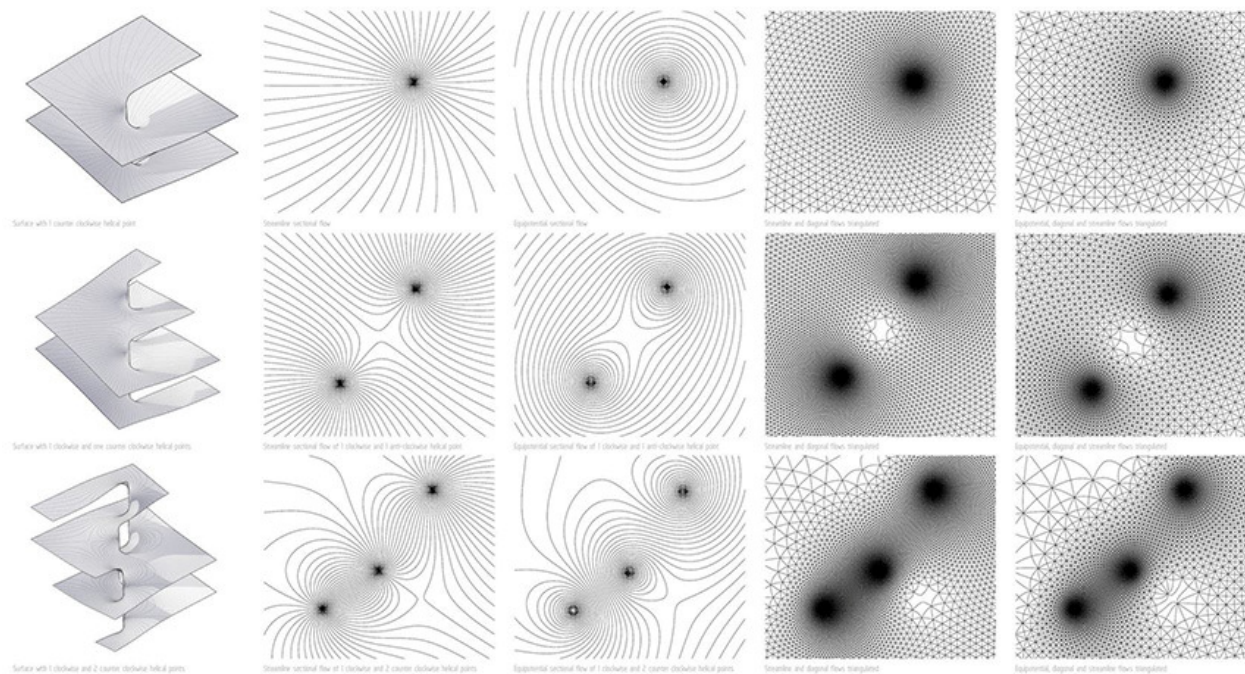
Due Date: Thursday, April 16th

Overview

Design a pavilion covering a 10x10 square foot area. The pavilion can take any mode (walkable, climbable, deployable, or other. Think outside of the “box” and explore innovative concepts and forms.

Objectives

- Gain experience with parametric thinking through sketching and modeling.
- Use interoperability tools to sync your design model into both documentation and rendering software.
- Evaluate the main components of rendering and understand the pros and cons of different platforms.



<https://spacesymmetrystructure.wordpress.com/rheotomic-surfaces/>

Assignment Tasks

1. Parametric Sketching

Begin with a conceptual sketch that defines geometric shapes using variables, constraints, and dimensions rather than fixed coordinates of your pavilion design. Layer parameter values, such as lengths, angles, or formulas visually as you sketch using different color pens or pencils. Sketches should convey your vision and serve as a foundation for your model.

2. Modeling

Use Rhino and Grasshopper to develop your pavilion design. Ensure your model incorporates the ideas presented in your initial sketch. Grasshopper definition must regenerate the primary geometry without manual post-processing steps. Document any required manual steps needed.

3. Documentation

Use RIR (Rhino-Inside-Revit) to bring your model into Revit. Create a plan, section, elevations, and a 3d view of your project. Include dimensions and tag important elements.

4. Rendering

Utilizing the detailed Revit model of your pavilion, render three images using the built in Twinmotion plugin for Revit and import your model into the Twinmotion rendering software. You are encouraged to pay special attention to the major components of photography such as lighting, composition, texture, and story.

Include three required images:

- **Bird's Eye View:** Capture the overall layout and design.
- **Interior View:** Showcase the interior space and experience, use a detail view if your pavilion does not have an "interior" space.
- **Exterior View:** Highlight the pavilion's relationship with its surroundings.

Submission Requirements

- Submit the following packaged into a zip file:
 - Initial sketches (scanned or photographed).
 - Rhino/Grasshopper model files (.3dm, .gh).
 - Revit model files (.rvt).
 - Rendered images (JPEG or PNG format).
 - Ensure all files are clearly labeled with your name and project title.
- A presentation pdf that includes documentation of your pavilion including sketches, grasshopper views, Revit drawings, diagrams, and renderings. This file will be separate from the zip file, but should be named the same.

Recommended Milestones

- **Week 1:** Concept sketch and program statement (1 paragraph).
- **Week 2:** Rhino + initial Grasshopper massing; test DirectShape export.
- **Week 3:** Revit model with family placement and parameter set.
- **Week 4:** Rendering test images; final refinements.

Final due: Thursday, April 16, 2026

Software Links / Additional Help

Installing Twinmotion: <https://www.twinmotion.com/en-US/download>

Rhino Inside Revit Help: <https://www.rhino3d.com/inside/revit/1.0/guides/>

Rhino / Grasshopper Help: <https://discourse.mcneel.com/>

In addition, you can schedule office hours at your convenience, and I will be happy to assist you and answer any questions not discussed in class.

Grading Criteria

Technical Skills (40%):

Demonstrates proficiency in Rhino, Grasshopper, and Revit with effective integration between platforms.

Rendering Quality (25%):

Produces high-quality visuals that clearly communicate the design intent with appropriate detail.

Presentation (15%):

Includes all required images, drawings, and sketches with clear and well-composed presentation.

Adherence to Guidelines (10%):

Meets all assignment requirements including timelines, file formats, and submission standards.

Creativity & Innovation (10%):

Shows originality in design concept and explores innovative approaches to materials and forms.

Grading Rubric (100 points total)

Criteria	Weight	Exemplary (A)	Proficient (B)	Satisfactory (C)	Needs Improvement (D-F)
Technical Skills	40%	Exceptional proficiency in Rhino, Grasshopper, and Revit; seamless integration.	Proficient use of software; minor issues in integration.	Basic skills demonstrated; some integration challenges.	Poor use of software; significant issues in integration.
Rendering Quality	25%	Stunning visuals; effectively communicates design intent with high detail.	Good quality renders; communicates design intent clearly.	Adequate renders; some detail lacking.	Poor quality renders; does not effectively communicate design.
Presentation	15%	All required images drawings and sketches are included and well-composed.	All elements included; minor composition issues.	Missing information; composition needs improvement.	Missing multiple pieces of information; poor composition.
Adherence to Guidelines	10%	All assignment requirements met with attention to detail.	Most requirements met; minor lapses.	Some requirements met; several lapses.	Many requirements not met; lacks overall coherence.
Creativity & Innovation	10%	Highly original concept; innovative use of materials and forms.	Original concept; some creative elements.	Basic ideas; limited exploration of concepts.	Lacks originality; minimal effort in design exploration.

Grading Scale

- **A:** 90-100
- **B:** 80-89
- **C:** 70-79
- **D:** 60-69
- **F:** 0-59